STRONG EFFECTIVENESS IN SQUARE-ZERO SYSTEMS IMPLIES OPENNESS OF VERSALITY

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 3281-3295 of sources/stacks-artin/pinned-original-source.tex; proof span is lines 3297-3384, the complete author proof (86 body lines).

Editorial note. Navigation stubs: 8 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 7 of queue_artin.json (pb=3812, stmt 3281-3295, proof 3297-3384).

Editorial note. stacks-artin wave ar01 (2026-10-14): fresh adjudication accept per lane-s3/artin_judgments.md; pool per Chief 10:5x direct dispatch (idle-first over S12's 01:5x note, SMC precedent); artin.tex 99 rows/67 with proof judged 46 accept / 21 reject / 5 AGP-overlap avoidance rows; patch-sd-envs contract spans scanned.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 045G: sources/stacks-artin/pinned-original-source.tex, source lines 3384–3384. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0CXM: sources/stacks-artin/pinned-original-source.tex, source lines 2013–2022. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0G2K: sources/stacks-artin/pinned-original-source.tex, source lines 3076–3084. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0G2S: sources/stacks-artin/pinned-original-source.tex, source lines 3176–3207. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0G2F: sources/stacks-artin/pinned-original-source.tex, source lines 3384–3384. Cited by the retained statement or proof but not part of

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the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0G2R: sources/stacks-artin/pinned-original-source.tex, source lines 3384–3384. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0CXT: sources/stacks-artin/pinned-original-source.tex, source lines 3257–3267. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 04AM: sources/stacks-artin/pinned-original-source.tex, source lines 3384–3384. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a locally Noetherian scheme. Let $\mathcal{X}$ be a category fibred in groupoids over $(Sch/S)_{fppf}$. Assume*

1. *$\Delta : \mathcal{X} \to \mathcal{X} \times \mathcal{X}$ is representable by algebraic spaces,*
2. *$\mathcal{X}$ has (RS*),*
3. *$\mathcal{X}$ is limit preserving,*
4. *systems (ξ_n) as in Remark 0CXT where $\mathrm{Ker}(R_m \to R_n)$ is an ideal of square zero for all $m \geq n$ are effective.*

Then $\mathcal{X}$ satisfies openness of versality.

Proof. Choose a scheme U locally of finite type over S, a finite type point u_0 of U, and an object x of $\mathcal{X}$ over U such that x is versal at u_0. After shrinking U we may assume U is affine and U maps into an affine open $\mathrm{Spec}(\Lambda)$ of S. Let $E \subset U$ be the set of finite type points u such that x is not versal at u. By Lemma 0G2K if $u \in E$ then u_0 is not a specialization of u. If openness of versality does not hold, then u_0 is in the closure $\overline{E}$ of E. By Properties, Lemma 0G2F we may choose a countable subset $E' \subset E$ with the same closure as E. By Properties, Lemma 0G2R we may assume there are no specializations among the points of E'. Observe that E' has to be (countably) infinite as u_0 isn't the specialization of any point of E' as pointed out above. Thus we can write $E' = \{u_1, u_2, u_3, \ldots\}$, there are no specializations among the u_i, and u_0 is in the closure of E'.

Choose $x \to x_1 \to x_2 \to \ldots$ lying over $U \to U_1 \to U_2 \to \ldots$ as in Lemma 0G2S. Write $U_n = \mathrm{Spec}(R_n)$ and $U = \mathrm{Spec}(R_0)$. Set $R = \lim R_n$. Observe that $R \to R_0$ is surjective with kernel an ideal of square zero. By assumption (4) we get ξ over $\mathrm{Spec}(R)$ whose base change to R_n is x_n. By assumption (3) we get that ξ comes from an object ξ' over $U' = \mathrm{Spec}(R')$ for some finite type Λ-subalgebra $R' \subset R$. After increasing R' we may and do assume that $R' \to R_0$ is surjective, so that $U \subset U'$ is a first order thickening. Thus we now have

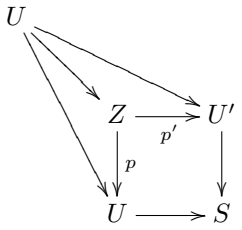
$$x \to x_1 \to x_2 \to \ldots \to \xi' \text{ lying over } U \to U_1 \to U_2 \to \ldots \to U'$$

By assumption (1) there is an algebraic space Z over S representing

$$(Sch/U)_{fppf} \times_{x,\mathcal{X},\xi'} (Sch/U')_{fppf}$$

see Algebraic Stacks, Lemma 045G. By construction of 2-fibre products, a T-valued point of Z corresponds to a triple (a, a', α) consisting of morphisms $a : T \to U$,

$a' : T \to U'$ and a morphism $\alpha : a^*x \to (a')^*\xi'$. We obtain a commutative diagram



The morphism $i : U \to Z$ comes the isomorphism $x \to \xi'|_U$. Let $z_0 = i(u_0) \in Z$. By Lemma 0CXM we see that $Z \to U'$ is smooth at z_0. After replacing U by an affine open neighbourhood of u_0, replacing U' by the corresponding open, and replacing Z by the intersection of the inverse images of these opens by p and p', we reach the situation where $Z \to U'$ is smooth along $i(U)$. Note that this also involves replacing u_n by a subsequence, namely by those indices such that u_n is in the open. Moreover, condition (3) of Lemma 0G2S is clearly preserved by shrinking U (all of the schemes U, U_n, U' have the same underlying topological space). Since $U \to U'$ is a first order thickening of affine schemes, we can choose a morphism $i' : U' \to Z$ such that $p' \circ i' = \mathrm{id}_{U'}$ and whose restriction to U is i (More on Morphisms of Spaces, Lemma 04AM). Pulling back the universal morphism $p^*x \to (p')^*\xi'$ by i' we obtain a morphism

$$\xi' \to x$$

lying over $p \circ i' : U' \to U$ such that the composition

$$x \to \xi' \to x$$

is the identity. Recall that we have $x_1 \to \xi'$ lying over the morphism $U_1 \to U'$. Composing we get a morphism $x_1 \to x$ whose existence contradicts condition (3) of Lemma 0G2S. This contradiction finishes the proof. □